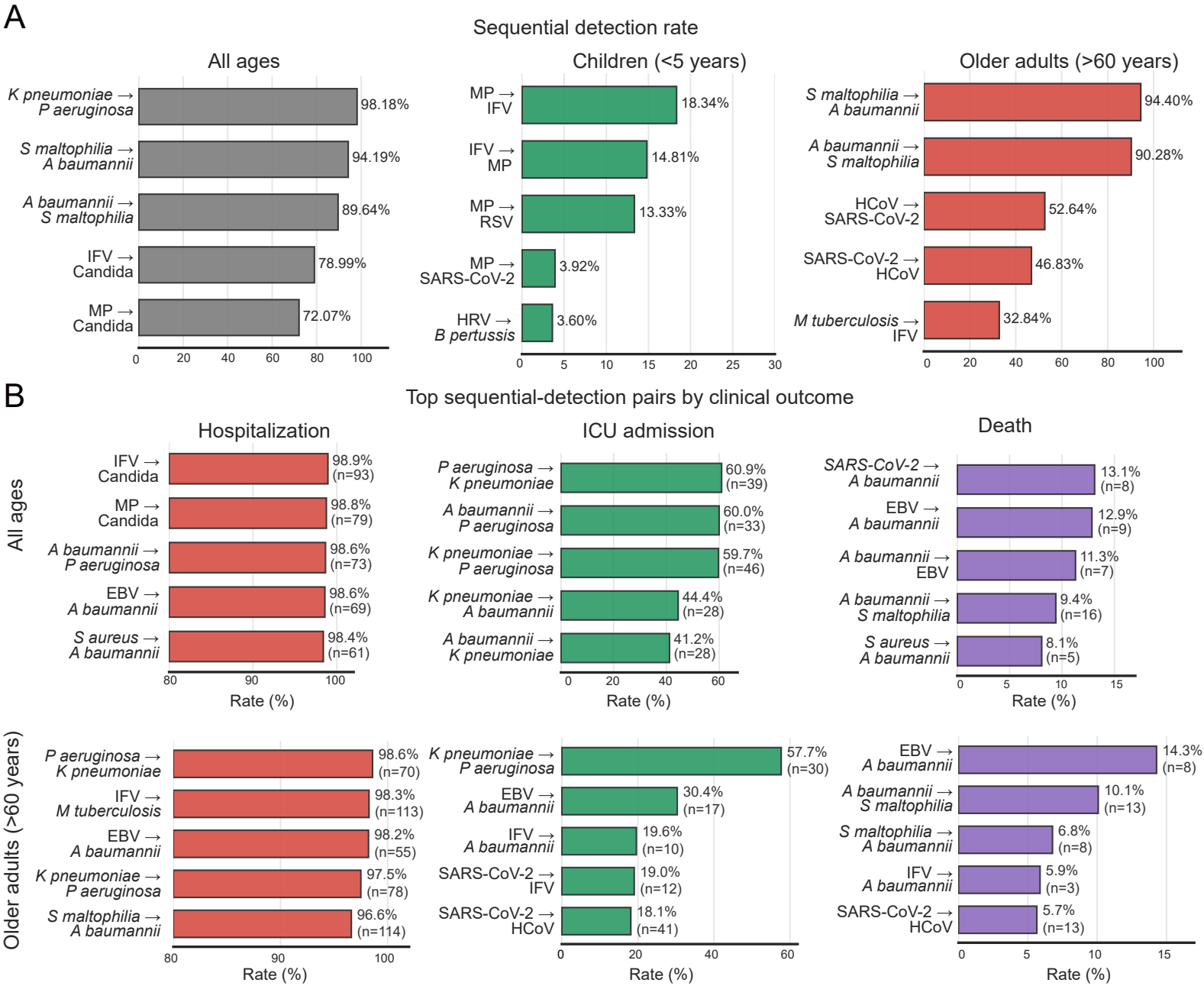

A
Sequential detection rate
All ages
K pneumoniae → P aeruginosa
98.18%
S maltophilia → A baumannii
94.19%
A baumannii → S maltophilia
89.64%
IFV → Candida
78.99%
MP → Candida
72.07%
0 20 40 60 80 100
Children (<5 years)
MP → IFV
18.34%
IFV → MP
14.81%
MP → RSV
13.33%
MP → SARS-CoV-2
3.92%
HRV → B pertussis
3.60%
0 5 10 15 20 25 30
Older adults (>60 years)
S maltophilia → A baumannii
94.40%
A baumannii → S maltophilia
90.28%
HCoV → SARS-CoV-2
52.64%
SARS-CoV-2 → HCoV
46.83%
M tuberculosis → IFV
32.84%
0 20 40 60 80 100
B
Top sequential-detection pairs by clinical outcome
Hospitalization
ICU admission
Death
All ages
IFV → Candida
98.9% (n=93)
MP → Candida
98.8% (n=79)
A baumannii → P aeruginosa
98.6% (n=73)
EBV → A baumannii
98.6% (n=69)
S aureus → A baumannii
98.4% (n=61)
80 90 100
Rate (%)
P aeruginosa → K pneumoniae
60.9% (n=39)
A baumannii → P aeruginosa
60.0% (n=33)
K pneumoniae → P aeruginosa
59.7% (n=46)
K pneumoniae → A baumannii
44.4% (n=28)
A baumannii → K pneumoniae
41.2% (n=28)
0 20 40 60
Rate (%)
SARS-CoV-2 → A baumannii
13.1% (n=8)
EBV → A baumannii
12.9% (n=9)
A baumannii → EBV
11.3% (n=7)
A baumannii → S maltophilia
9.4% (n=16)
S aureus → A baumannii
8.1% (n=5)
0 5 10 15
Rate (%)
Older adults (>60 years)
P aeruginosa → K pneumoniae
98.6% (n=70)
IFV → M tuberculosis
98.3% (n=113)
EBV → A baumannii
98.2% (n=55)
K pneumoniae → P aeruginosa
97.5% (n=78)
S maltophilia → A baumannii
96.6% (n=114)
80 90 100
Rate (%)
K pneumoniae → P aeruginosa
57.7% (n=30)
EBV → A baumannii
30.4% (n=17)
IFV → A baumannii
19.6% (n=10)
SARS-CoV-2 → IFV
19.0% (n=12)
SARS-CoV-2 → HCoV
18.1% (n=41)
0 20 40 60
Rate (%)
EBV → A baumannii
14.3% (n=8)
A baumannii → S maltophilia
10.1% (n=13)
S maltophilia → A baumannii
6.8% (n=8)
IFV → A baumannii
5.9% (n=3)
SARS-CoV-2 → HCoV
5.7% (n=13)
0 5 10 15
Rate (%)

Figure 5